Senior+ Engineering for Forward Deployed Engineers

Building Systems That Survive Customers, Scale, and Scrutiny

What an FDE Is (Set Context Early)

Forward Deployed Engineer = Engineer + Consultant + Owner

An FDE is judged on:

→ Can the customer **run this without you?**

→ Does it **break under real usage?**

→ Can it **scale beyond the POC?**

If the customer depends on you to explain or fix it, it's not Senior+.

Why This Checklist Exists

Most Customer Projects Fail Here

→ Works in demo

→ Fails in real usage

→ No observability

→ No fallback

→ No documentation

Senior FDEs reduce customer risk.

How Customers Review Your Work

The Unspoken Evaluation

Customers don’t ask:

Is this elegant?

Customers ask:

What happens when this breaks?

Can my team own this?

Can this go to production?

The 6 Senior+ Dimensions (FDE View)

1. Filesystem & Documentation
2. Architecture & Design Integrity
3. Implementation & Code Quality
4. Testing & Validation
5. Reviewer / Customer Lens
6. Final Production Benchmark

Section 1: Filesystem & Documentation

Customer Reality

→ They won’t read your mind

→ They will inherit your system

Mandatory for Customer Handoff

/requirements → customer constraints

/docs/architecture → system ownership map

/docs/data-flow → how data actually moves

/src → clean service boundaries

/tests → proof, not promises

README.md → onboarding guide

README ≠ Code → **customer escalation**

Section 2: Architecture vs Data Flow

Architecture	Data Flow
Who owns what	What happens when
Infra view	Logic view

Section 2: POC vs Production (Customer Lens)

This Is Where FDEs Get Burned

POC

Single service

Manual steps

No monitoring

Production

Gateway

Scaling

Alerts

Runbooks

“Customers don’t want demos.
They want survivability.”

Section 2: Design Decisions (ADRs)

Customers Ask “Why?”

Why this DB?

Why this model?

Why this cost?

Senior FDE Answer:
Trade-offs, not opinions.

- An **Architecture Decision Record (ADR)** is a **short, structured document** that captures **why** a significant technical or architectural decision was made, **what was decided**, and **what alternatives were considered**.

Section 3: Implementation & Code Quality

Production Code = Failure-Aware Code

Immediate Red Flags for Customers

- print logs
- hardcoded keys
- single massive file
- no validation

Section 3: FDE Code Signals

- Structured logs → supportability
- Validation → data trust
- Pooling → performance under load
- Docker → customer reproducibility
- “If the customer can’t run it, it’s not delivered.”

Section 4: Testing & Validation

- **“Looks Good” Is Not a Metric**
- Customers ask:
- Is it accurate?
- Is it fast?
- Is it stable?

Section 4: Senior FDE Validation

- Ground truth dataset
- Automated evaluation
- Metrics:
  - Latency (p95/p99)
  - Accuracy (NDCG / LLM-as-Judge)
- **No metrics = no credibility**

Section 5: Reviewer / Customer Lens

You Are Being Scanned, Not Studied

They scan for:

- Failure handling
- Observability
- Scalability
- Ownership clarity

Section 6: What “YES” Means

Senior+ FDE Benchmark

Stateless

Scalable

Observable

Documented

Transferable

“If I leave tomorrow, this still runs.”